# Wildfire Exposure Assessment For Algerian Barbary Macaques (*Macaca sylvanus*)

## A spatial assesment of Barbary Macaque habitat overlap with 2026 wildfires in Northeastern Algeria

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## Introduction

The Barbary Macaque (*Macaca sylvanus*) is North Africa's only primate and has been classified as endangered by the IUCN Red list as of 2008. The species occupies a small and heavily fragmented range within forested regions of Morocco and Algeria, along with a small introduced population on Rock of Gibraltar. It faces critical threats from deforestation, urbanization, agriculture, and human disturbances driven primarily by tourism.



Figure 1: Barbary Macaques pictured in the Chiffa Gorge in Blida, Algeria. Photo: Louiza Mohamed-Cherif

Algeria holds nearly 25% of the Barbary Macaque's global population, however there is a crucial need for new reviews as a comprehensive census of the existing populations within the country does not exist (Boumenir et al ., 2025) Even more pressing, recent dry seasons driven by climate change have worsened regional wild fires, which overlap significantly with what remains of Algerian macaque habitat.



Figure 2: A wildfire across a forested mountain in Northeast Algeria. Photo: Hafid Derradji/ Facebook via Morocco World News

In the summer of 2026 nearly 160 wildfires broke out within a 24 hour period, stretching along the coastline of Algeria. These fires were felt across the north-eastern, central, and western portions of the country, with the highest intensity occurring in the north-east. The rapid scale and severity of these fires within such a short time frame signifies an urgent need to asses their impact on vulnerable wildlife habitat.

This study aims to map the spatial intersection of 2026 wildfire damage with known macaque habitat to estimate recent habitat loss. Wildfire damages intersect with protected areas will also be evaluated, to determine whether existing conservation frameworks are effectively safeguarding the Barbary Macaque.

## Methods

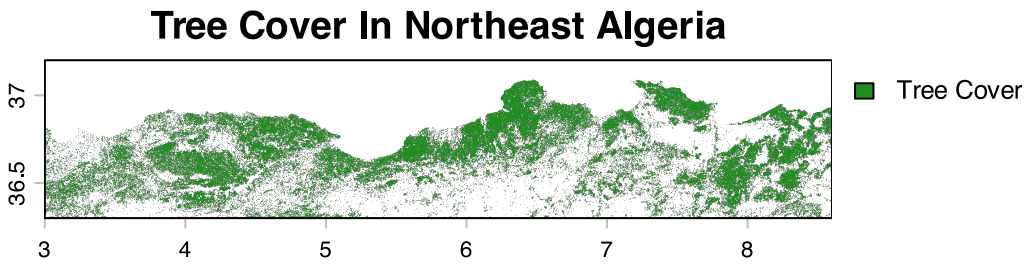
1. The area of interest was first defined as a region of approximately 50,028.83km², surrounding the forested coastline of northeast Algeria. The AOI contains about 2092km² of protected land; It encompasses nearly the entirity of Algerian macaque habitat, however excludes a singular small patch located in Blida (a few miles southwest of Algiers, falling outside the region of highest fire concentration).

2. Relevant data sets pertaining to land cover class, burned area, species range, known protected land, and occurrence (?) were cleaned and cropped to a bounding box within the study area.
3. Suitable habitat range was defined as areas where tree-cover intersects with known habitat range polygons. This represented a total area of about 648~ km^2 within the study region.
4. The overlapping area between defined suitable habitat and burned area polygons was calculated using st_intersection(), the total square km of habitat exposed to fire was then calculated using st_area().
5. Intersect with burned and area was calculated for protected areas.
6. known species occurrence points (?)

## Data:

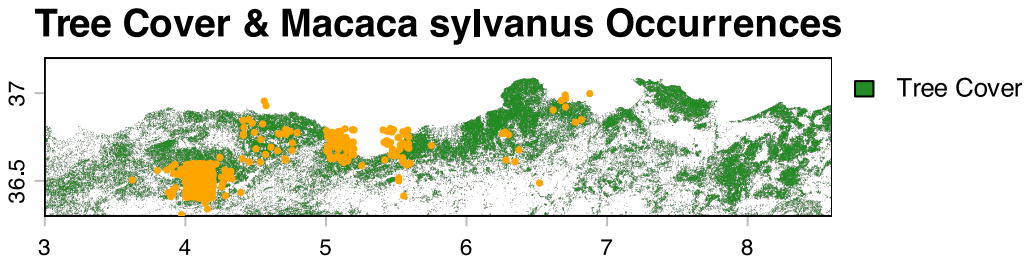
### 1. ESA World Cover For Tree Cover Data

Land cover data downloaded from ESA World Cover from the year 2021, masked to Tree Cover.



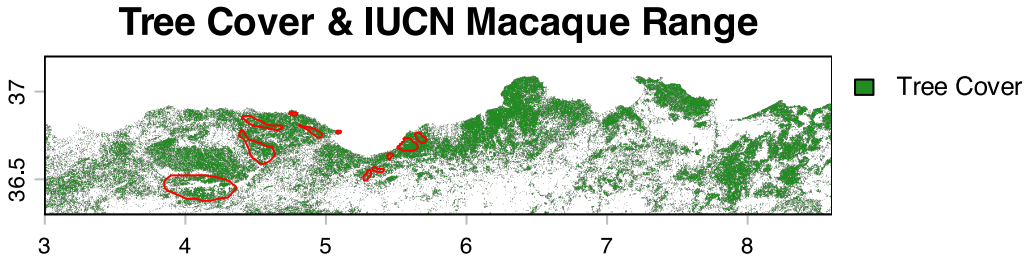

### 2. Species Occurrence Points

Occurrence points collected from the Global Biodiversity Information Facility (gbif) - may be excluded from final- see notes at end of page.



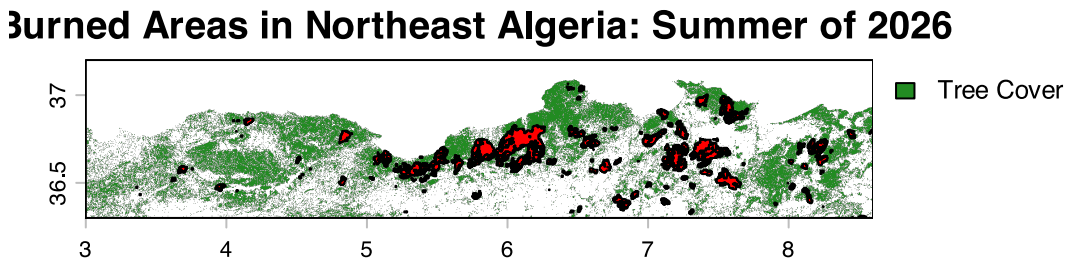

### 3. Species Range Map

Habitat range as defined by the IUCN red list.



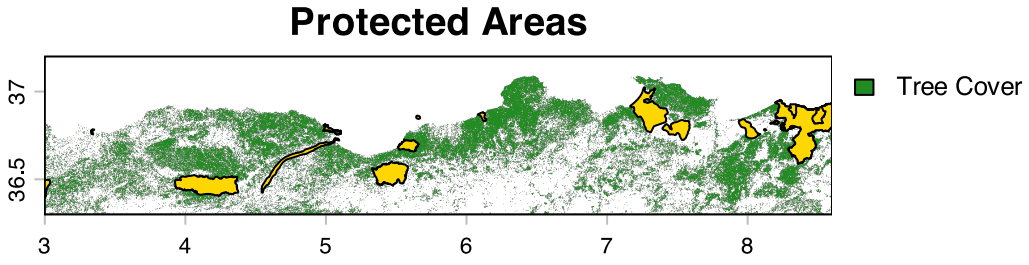

### 4. Burned Area Data

VIIRS/NPP Burned Area Monthly L4 Global 500m SIN Grid V002, from 1 June 2026 to 31 October 2026 (Date range stretched to include the entire summer fire season).



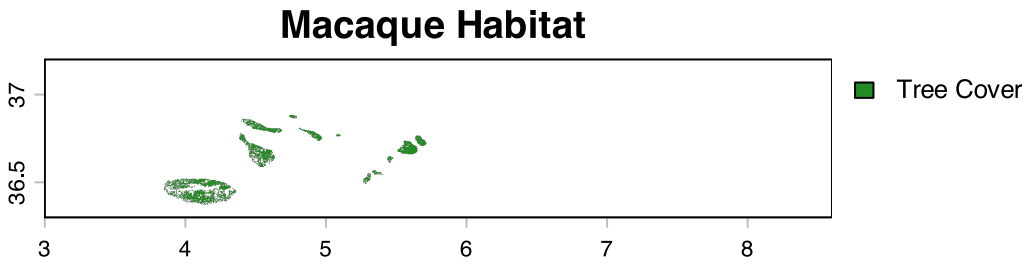

## 5. Protected Areas

World Database on Protected and Conserved Areas (WDPA), filtered to most recent data set (2026).



# Data Processing & Analysis

## 1. Creating a suitable habitat raster




## 2. Calculating total habitat burned

```
burned_sf <- st_make_valid(burned_sf)
habitat_sf <- st_make_valid(habitat_sf)
#1. Calculate the overlap of burned area and macaque habitat
burned_hab_union <- st_union(burned_sf) # 1.1 combine burned area into one
combined total
burned_hab_union <- st_make_valid(burned_hab_union)
# 1.2 intersect burned area with habitat
burned_hab <- st_intersection(burned_hab_union, habitat_sf)
# 1.2.1 - check by running plot(burned_hab)
# 1.3 calculate total area burned
burned_area <- burned_hab |>
  st_as_sf() |>  # 1.3.1 convert burned_hab back into an sf before passing to
st_area
  mutate(burned_area = st_area(x)) # 1.3.2 x notates the single column in
burned_hab

# burned_area is approx. 40.5km2
# 1.4 find percent habitat burned

percent_burned <- (as.numeric(burned_area$burned_area)/
as.numeric(hab_area))*100
```

```
# percent_burned -- they lost 6.23% of habitat in the northeast in ONE SUMMER!
That's not good!
```

### 3. Calculating protected areas burned

1. plan to –> find intersect of protected areas and habitat (how much of macaque habitat actually falls in protected areas)
2. find total area of protected habitat burned (follow work flow from step 2)
3. find how much of protected area intersect with actual macaque habitat was burned

### 4. validation with species occurrence points

Notes: did not specify year when pulling occurrence data from GBIF, I'm going to try filtering to recent years first- if it does not return enough points for validation it may be better to cut species points out of the study rather then use outdated occurrences, but I have not decided yet.

## Results

1. table - want to show how much habitat macaques have in study area - how much was burned, how much habitat they have in protected areas - how much of that was burned
2. Final map/visual with all data layered

## Conclusions

## Limitations

## References